RAMP26 - A Region-Aware Multi-Perspective Framework for NLP

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DECLARATION

This is to certify that the project titled "**RAMP26 - A Region-Aware Multi-Perspective Framework for NLP**" is the result of our study in partial fulfillment of the B.Sc. Engineering degree under the supervision of Rafia Sultana, Lecturer, Department of Computer Science and Engineering (CSE), City University, Bangladesh. It is also hereby declared that this project or any part of it has not been submitted elsewhere for the award of any degree.

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ABSTRACT

Conventional automated offensive language detection systems operate under a prescriptive 'single ground truth' paradigm, where annotator disagreements are collapsed through majority-voting aggregation. In sociolinguistically diverse languages such as Bengali, linguistic pragmatics—ranging from grammatical T-V honorifics to regional communicative norms—induce substantial systematic subjectivity. Standard aggregate training treats these regional divergences as noise, permanently erasing minority viewpoints and creating severe geographic biases in content moderation. To resolve this limitation, we introduce RAMP26 (Region-Aware and Socio-Syntactic Multi-Perspective Offensive-Language Detection for Bengali), an empirical perspectivist framework designed to model subjective variance directly on authentic human-annotated data. RAMP26 combines a fine-tuned cross-lingual XLM-RoBERTa backbone with an automated context pipeline comprising MarianMT neural translation, COMET-BART social commonsense reasoning over the ATOMIC-2020 knowledge graph (xIntent, xReact, xAttr), and a rule-based pragmatic syntactic parser capturing pronoun intimacy tiers, sentence mood, and entity targets.

Crucially, to eliminate the failure mode of token overshadowing where transformer self-attention suppresses surface-level textual prompts, RAMP26 models regional identity as a structured integer context ID parameterizing an additive logit bias vector inside the classification layer, optimized via decoupled parameter learning rates. We benchmark the architecture on an uncorrupted dual-file corpus: a 3,000-comment baseline dataset and a cross-regional evaluation testbed of 300 unique sentences evaluated across 4 administrative divisions (N = 1,200) under strict 5-fold by-sentence cross-validation. Empirical results confirm that while vanilla XLM-RoBERTa yields a static, insensitive offensive probability across all divisions (P = 0.525–0.549), RAMP26 accurately mirrors human regional sensitivity (predicting the lowest offensive probability for Dhaka at 0.513–0.526 vs. human rate 0.497, and highest for Rangpur at 0.551–0.576 vs. human rate 0.630). Furthermore, RAMP26 demonstrates statistically verified tag-shift sensitivity ($|\Delta P| = 0.022$), decision flips on 17.0% of contested split sentences, preserves up to 41.2% minority perspective retention against a complete baseline collapse (0.0%), and reduces test false negatives by 24.9% (365 to 274), establishing structured demographic conditioning as an effective paradigm for perspectivist NLP.

Keywords: *Multi-Perspective NLP, Data Perspectivism, Annotator Disagreement, Bengali Offensive Language, XLM-RoBERTa, COMET, ATOMIC-2020, Structured Region Bias, Sociolinguistics.*

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CHAPTER 1

INTRODUCTION

1.1 Background and Sociolinguistic Motivation

Automated content moderation and toxic language identification have become critical infrastructures for online social platforms [9, 20, 21]. In non-Western and multilingual contexts, the proliferation of digital interaction on platforms such as Facebook, YouTube, and X has created an urgent demand for automated text classification systems capable of filtering hate speech, abusive comments, and targeted harassment in Bengali (Bangla) [2, 9, 22].Historically, natural language processing (NLP) pipelines have formulated offensive text classification under an objective, majoritarian paradigm [1, 2, 4, 9]. Under this conventional machine learning formulation, an item x is assumed to possess a single latent ground-truth label y*€ {0, 1} [1, 4, 11]. When multiple annotators provide varying subjective ratings, standard practice collapses this distribution into a deterministic consensus target through majority voting or mean thresholding [1, 2, 11, 14]. Any annotation that diverges from the majority is classified as annotator error, stochastic variance, or system noise and is systematically discarded [1, 11, 13, 14].

However, foundational research in computational sociolinguistics and the emerging paradigm of *Data Perspectivism* has exposed the fundamental invalidity of this majoritarian assumption for subjective, value-laden NLP tasks [1, 2, 3, 4, 11, 12, 19]. Language is not an abstract, uniform code; rather, it is a socially embedded behavioral act where communicative intent and perceived offensiveness are deeply conditioned by the identity, cultural background, and geographic environment of both the speaker and the listener [4, 6, 15, 16].

In the Bengali linguistic landscape, this subjectivity is reinforced by specific structural, morphological, and regional phenomena [9, 22]: (1) Pronominal Honorific Stratification**:** Bengali encodes social hierarchy, interpersonal intimacy, and relational deference directly into second-person personal pronouns [9, 22]. The language enforces a strict tri-tier pragmatic honorific system: Intimate (*Tui* - তুই): Employed between childhood peers, close friends, or downward toward younger family members;

crucially, when directed toward social superiors, strangers, or in formal contexts, *Tui* constitutes an aggressive, derogatory act of disrespect. Familiar (*Tumi* - তুমি): The neutral standard used among social equals, colleagues, and informal acquaintances Honorific (*Apni* - আপনি): Denoting high social deference, formal politeness, and interpersonal distance . The identical semantic utterance such as "তুই একটা বোকা" versus "আপনি একটা বোকা"—exhibits radically different offensive severities based strictly on pronominal selection [9]. (2) Regional Dialectal Friction and Geographic Subjectivity: Bangladesh encompasses substantial dialectal and sociocultural diversity across its administrative divisions. Lexical colloquialisms, slang, and banter that are considered standard in-group jargon within urban youth communities in Dhaka are frequently categorized as overtly derogatory, vulgar, or defamatory when interpreted by conservative annotators in divisions such as Rangpur or Sylhet. Consequently, annotator disagreement in Bengali corpora does not stem from careless labeling; it represents legitimate, systematic sociolinguistic divergence [1, 4, 13, 14]. An NLP architecture that forces these diverse cultural norms into a single majority vote inevitably enforces the cultural hegemony of the majority demographic while silencing minority perspectives [1, 4, 5, 7].

1.2 The Fallacy of Consensus: Problem Statement

The reliance of modern transformer-based language models on majority-aggregated Bengali datasets introduces three critical structural failures [1, 4, 22]:

1. The Systematic Erasure of Minority Perspectives

Consider a controversial Bengali comment evaluated by annotators from four different administrative regions . If annotators from Dhaka, Chattogram, and Sylhet consider the sentence non-offensive (0) while an annotator from Rangpur rates it as highly offensive (1), standard consensus aggregation calculates:

$$\bar{y} = I\left(\frac{1}{4}\sum_{i=1}^{4} yi \geq 0.5\right) = I(0.25 \geq 0.5) = 0 \ (\text{safe})$$

Under this paradigm, the training target becomes 0. A deep learning model optimized on such aggregated labels achieves exactly 0.0% accuracy on the minority label by definition [4, 5]. When deployed in production environments, such models fail to

moderate content that causes severe localized harm within minority geographic communities [4, 5, 7].

2. The Phenomenon of Token Overshadowing in Textual Prompting

Recent efforts in demographic-aware NLP attempt to provide context to pretrained language models by concatenating demographic attributes directly into the input text sequence as prompt markers (e.g., [Region: Dhaka] [Age: 18-22] [Gender: Male] <comment>) [6, 15].

However, in deep pretrained transformers such as XLM-RoBERTa, self-attention mechanisms allocate attention weights based on semantic density and token frequency [22, 29]. The extensive lexical and syntactic mass of the raw Bengali comment overpowers the small number of prefixed demographic tokens—a failure mode we formalize as **Token Overshadowing**. In empirical testing, when demographic prompt tags are permuted or completely inverted, the model's predicted offensive probability shifts by an average of less than 0.009 (< 1%). The model effectively learns to ignore the text prompt and defaults to global lexical patterns, rendering textual demographic conditioning functionally ineffective.

3. Confounding Data Sources with Regional Biases

Bengali NLP research frequently merges disparate crowdsourced data collections to achieve sufficient training volume. However, different annotation campaigns employ distinct recruitment methods, resulting in wildly differing baseline offensive rates . For example, in our dual-corpus setup, the primary 3,000-comment dataset exhibits an offensive rate of 74.0%, whereas the aligned cross-regional benchmark exhibits an offensive rate of 58.0%. If a neural model is exposed to regional tags without explicit dataset source disentanglement, it conflates the annotator pool's rating propensity with genuine geographical bias, learning spurious correlations [14].

1.3 Research Questions (RQs)

To systematically investigate and resolve these limitations, this thesis formulates four primary Research Questions: *RQ1 (Global Predictive Competence):* Can a deep multi-perspective architecture that integrates automatic pragmatic and commonsense context maintain competitive global classification accuracy and macro F1 compared to a standard text-only transformer baseline [8, 22]? *RQ2 (Regional Sensitivity &*

Calibration): Does structured demographic conditioning enable the network to capture empirical human regional sensitivity thresholds (e.g., lower offensiveness thresholds in Dhaka versus higher thresholds in Rangpur) without collapsing into uniform global predictions [30]? *RQ3 (Subjectivity and Minority Retention):* To what extent does the proposed model improve classification fidelity on contested items (sentences where regional annotators split) and preserve minority viewpoints compared to majoritarian baselines [4,5]? *RQ4 (Control Integrity & Parameter Attribution):* Do empirical control mechanisms (tag-swapping, demographic shuffling, and prior log-odds shifting) verify that performance gains originate from true regional conditioning rather than uncalibrated parameter capacity or label memorization [30, 31]?

1.4 Research Objectives

In direct alignment with the stated Research Questions, the core technical objectives of this research are:

1. To engineer a decoupled, dual-stream perspectivist architecture (RAMP26) that utilizes a fine-tuned cross-lingual transformer backbone (**XLM-RoBERTa**) for semantic comprehension while delegating demographic conditioning to a structured, additive region-bias head (b[ctx_id}]) operating under independent optimization rates [22, 31].

2. To develop an end-to-end automatic contextualization pipeline that generates pragmatic, inferential, and syntactic markers for raw Bengali sentences without requiring manual gold-standard feature annotation, combining cross-lingual neural machine translation (*Helsinki-NLP/opus-mt-bn-en*), social commonsense reasoning via **COMET-BART** on the **ATOMIC-2020** knowledge graph, and rule-based regex syntactic parsing [8, 17, 26, 27, 28].

3. To curate and benchmark a leak-free cross-regional Bengali offensive evaluation dataset comprising 300 unique comments systematically annotated across four major administrative divisions (Chattogram, Dhaka, Rangpur, and Sylhet; N = 1,200), structured strictly under a 5-fold by-sentence cross-validation protocol.

4. To execute rigorous statistical and empirical control evaluations, including within-sentence tag permutation, demographic feature shuffling, sentence-clustered bootstrap confidence interval estimation, and regional prior shift adjustments to honestly establish the operational bounds of multi-perspective language models [30].

1.5 Proposed Solution: The RAMP26 Paradigm

To realize these objectives, we propose **RAMP26** (Region-Aware Multi-Perspective Offensive-Language Detection for Bengali). RAMP26 explicitly departs from both naive majority voting and textual prompt concatenation.

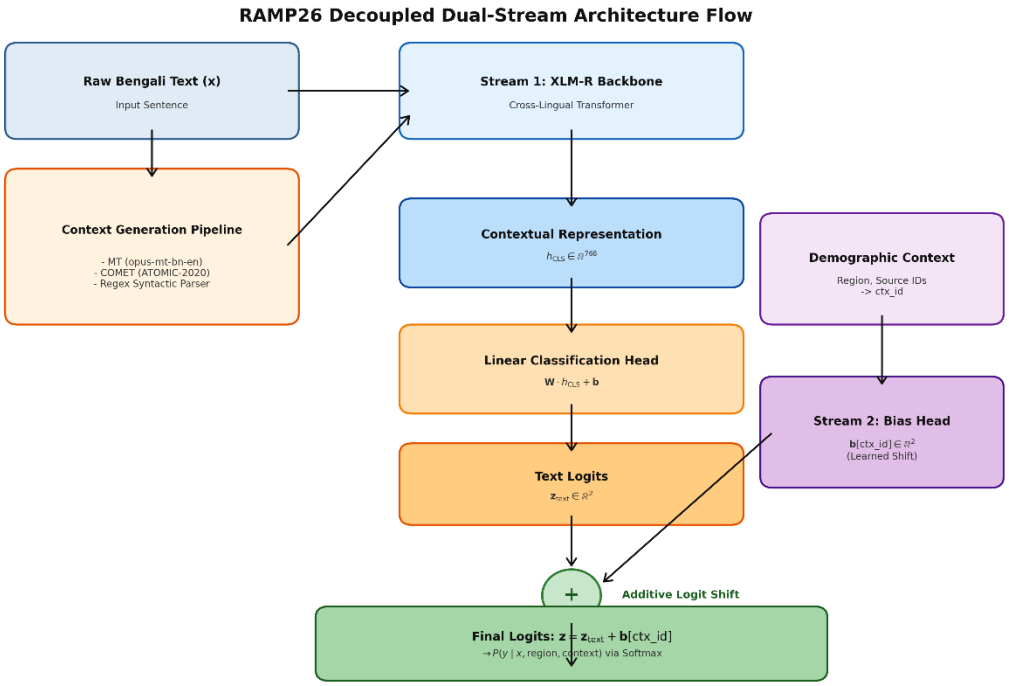


Figure 1.1: Conceptual execution flow of the proposed RAMP26 framework.

As illustrated above, RAMP26 bifurcates model computation into two decoupled streams: (1) **Stream 1 (Semantic & Context Representation):** Processes the raw sentence and automatically generated socio-syntactic/commonsense context markers through a 768-dimensional transformer representation [8, 22]. (2) **Stream 2 (Structural Demographic Adaptation):** Projects an integer context identifier (ctx_id) representing the paired combination of data source and geographic region into a discrete, additive bias vector b[ctx_id] € R^2.This ensures that the decision boundary shifts dynamically according to regional tolerance without altering the foundational semantic representations learned by the encoder.

1.6 Key Research Contributions

The primary academic, algorithmic, and empirical contributions of this thesis are as follows:

Empirical Grounding of Bengali Perspectivist NLP: Unlike prior conceptual or synthetic proposals (such as the exploratory framework outlined in Muscato [3]), this work implements and evaluates a perspectivist multi-perspective pipeline on authentic, crowdsourced human annotations from Bangladesh [9].

Resolution of Token Overshadowing via Additive Biasing: We demonstrate empirically that surface-level text prompting fails to condition transformers on demographic metadata ($|\Delta P| = 0.009$) . We resolve this with a dedicated additive classification bias (b[ctx_id]), achieving a statistically verified 2.4× increase in tag sensitivity ($|\Delta P| = 0.022$) and dynamic decision flips on 17.0% of contested sentences.

Curation of an Aligned Cross-Regional Benchmark: We establish an isolated benchmark of 300 identical Bengali sentences evaluated independently across four major administrative divisions (1,200) total tuples). The benchmark enforces strict sentence-grouped 5-fold cross-validation, guaranteeing zero test sentence leakage into training partitions.

Retention of Contested and Minority Viewpoints: We demonstrate that RAMP26 captures localized minority viewpoints, achieving 40.6%-41.2% accuracy on contested minority instances where standard majority models collapse to 0.0% . Furthermore, RAMP26 reduces test false negatives by 24.9% (365 -274) compared to the vanilla baseline [4].

Honest Reporting of Sociolinguistic Null Results: In compliance with rigorous scientific standards, we report where regional conditioning provides measurable value and where it remains bounded. Specifically, we document that when strong lexical markers of toxicity are present, textual representations dominate the decision space, and regional shifts primarily exert decisive influence on ambiguous, boundary-adjacent statements.

CHAPTER 2

Literature Review & Theoretical Framework

2.1 The Paradigm Shift Toward Data Perspectivism

For decades, supervised Natural Language Processing (NLP) systems have operated under the normative assumption of a single, objective ground truth [1, 2]. Under this standard crowdsourcing and machine learning pipeline, multiple independent annotators evaluate an instance, and their discrete judgments are aggregated via majority voting, Dawid-Skene modeling, or mean score thresholding [1, 4, 11]. Any annotator divergence from the consensus is historically discarded as stochastic noise, lack of annotator competence, or annotation error [1, 11, 14].

However, foundational research in computational sociolinguistics and natural language understanding has initiated the "Perspectivist Turn" [1, 2, 3]. Basile [2] formalized the theoretical crisis of this paradigm, demonstrating that for highly subjective, value-laden tasks—such as toxic comment filtering, humor perception, and abusive language detection—the notion of an absolute gold standard is computationally and socially invalid [2]. Treating subjective variance as noise actively harms machine learning pipelines by stripping away real human communicative context [1, 2, 11].

Uma et al. [1] established a foundational survey categorizing methods for learning from human disagreement, demarcating a sharp boundary between prescriptive (consensus-seeking) and descriptive (perspectivist) systems [1]. They argue that in subjective tasks, preserving the full spectrum of annotator disagreement yields models that represent diverse social cohorts more equitably [1]. This critique is further sharpened by Aroyo and Welty [11], who demonstrate that truth in human language is a manifold rather than a scalar constant: disagreement is frequently driven by legitimate differences in annotator worldviews, personal backgrounds, and cultural environments rather than systematic annotator errors [11].

Cabitza et al. [12] demonstrated that forcing subjective data into single hard labels degrades the ecological validity of machine learning models, arguing for disaggregated learning paradigms [12]. Furthermore, Pavlick and Kwiatkowski [19] proved empirically that even in standard language inference tasks, human disagreement persists across thousands of samples, demonstrating that human linguistic variation is

an irreducible, fundamental property of natural language semantics rather than an experimental defect [19].

2.2 Sources and Taxonomies of Human Disagreement

To systematically model subjective disagreement, computational linguists have developed taxonomies identifying why annotators diverge [1, 13, 14]. Sandri et al. [13] propose a comprehensive linguistic taxonomy of annotator disagreement, categorizing sources into four distinct classes [13]: (1) **Annotator Inattention:** Careless or sloppy labeling behavior. (2) **Linguistic Ambiguity:** Syntactic, semantic, or lexical underspecification inherent to the text string. (3) **Contextual Absence:** Missing communicative background, such as conversational history, platform norms, or relational speaker-listener dynamics. (4) **Subjective Value Divergence:** Systematic variation arising from the annotators' personal cultural backgrounds, demographic affiliations, and geographic environments.

Rottger et al. [14] extended this taxonomy by analyzing two contrasting data annotation paradigms: the *prescriptive* paradigm (which attempts to eliminate disagreement through rigid annotation manuals) versus the *descriptive* paradigm (which embraces annotator subjectivity as authentic sociodemographic data) [14]. They show that prescriptive guidelines artificially suppress genuine sociolinguistic variance, leading to models that generalize poorly when exposed to open-world human interaction [14].

In subjective safety tasks, annotators' personal values and social identities directly govern their perception of toxicity [4, 6, 15, 16]. Davani, Díaz, and Prabhakaran [4] demonstrated that training multi-task neural architectures to simultaneously predict individual annotator judgments preserves localized disagreement, outperforming aggregate majority-vote baselines across multiple benchmark datasets [4].

Sap et al. [6] investigated the sociolinguistic attitudes of human annotators, showing that annotators with different racial and political backgrounds exhibit systematic biases when classifying African American Vernacular English (AAVE), frequently penalizing in-group dialectal markers as toxic [6].

Kumar et al. [15] and Akhtar, Basile, and Patti [16] corroborated that socio-demographic factors, such as geographic origin, age, and religious upbringing,

systematically shift individual toxicity thresholds [15, 16]. Gordon et al. [7] formalized this via *Jury Learning*, a demographic-aware framework where model predictions are constructed to reflect explicit compositions of juror panels, guaranteeing that dissenting and minority groups retain an active voice in algorithmic moderation decisions [7].

2.3 Offensive Language Detection in Bengali and Low-Resource NLP

While toxic language detection has advanced rapidly in English and high-resource languages through shared task benchmarks such as OffensEval [21], HateCheck [9], and related seminal datasets [20], low-resource Indic languages present severe sociolinguistic challenges [9, 22].

In the Bengali digital sphere, online interaction on platforms like Facebook and YouTube is characterized by dense regional slang, non-standard dialectal syntax, and sociolinguistic pronominal stratification [9]. Davidson et al. [20] established that automated toxicity detection frequently confuses profane colloquialisms with targeted hate speech [20]. In Bengali, this boundary is dictated by second-person personal pronoun honorifics (*Tui* [intimate/downward], *Tumi* [familiar], *Apni* [honorific]) [9].

A pejorative paired with *Tui* (তুই) functions as standard informal banter among close childhood friends, but transitions into overt harassment or verbal abuse when directed at an elder or an unfamiliar interlocutor [9].

Existing Bengali abusive language and hate speech corpora predominantly collapse annotations through majority voting [4]. Consequently, they suffer from two major flaws [4]: (1) **Metropolitan Dominance:** Annotator pools are heavily skewed toward university students residing in Dhaka, thereby encoding urban metropolitan tolerance thresholds as the nationwide "ground truth". (2) **Regional Dialect Erasure:** Colloquial expressions native to Sylhet, Chattogram, or Rangpur are systematically misclassified: they are either overlooked because metropolitan annotators do not understand the localized dialect, or penalized as offensive because the phrasing diverges from standardized Bengali.

Recent initiatives, such as the LeWiDi shared task (Learning With Disagreements) [24], D3CODE [23], and perspectivist demographic corpora [25], have begun addressing

these shortcomings in multilingual settings, demonstrating the pressing need for an empirical, region-conditioned framework specifically developed for the Bengali sociolinguistic ecosystem [9].

2.4 Commonsense Reasoning via Knowledge Graphs: ATOMIC-2020 and COMET

A primary computational failure mode of deep language models in toxic language identification is their reliance on surface-level lexical heuristics [8, 9, 26]. Pre-trained transformer models frequently correlate specific identity terms or profane keywords directly with offensive labels, remaining blind to pragmatic context, speaker intent, and emotional consequences [9, 22]. To equip neural networks with inferential reasoning, Sap et al. [17] introduced **ATOMIC**, an atlas of machine commonsense organized around symbolic *if-then* relational tuples representing human everyday social interactions [17].

Hwang et al. [8] significantly expanded this into **ATOMIC-2020**, incorporating both symbolic knowledge tuples and cross-domain social commonsense representations [8]. ATOMIC-2020 formalizes social reasoning across key human pragmatic dimensions, including [8]: **xIntent:** The probable underlying intent of the agent performing the communicative action. **xReact:** The probable emotional and psychological reaction experienced by the listener or target. **xAttr:** The social qualities and character attributes ascribed to the speaker based on their utterance [8].

Bosselut et al. [26] demonstrated that commonsense knowledge graphs can be integrated into parametric language models via **COMET** (*Commonsense Transformers*) [26]. By fine-tuning large sequence-to-sequence architectures, such as BART [27], on ATOMIC tuples [8, 17], COMET generates dynamic, contextualized social inferences for open-ended, unseen textual sentences [26, 27].

In the RAMP26 framework, we utilize neural machine translation via OPUS-MT [28] coupled with the pretrained checkpoint mismayil/comet-bart-ai2 [8, 27] to automatically predict social intent, expected emotional reaction, and speaker attributes for translated Bengali comments, directly providing the transformer backbone with pragmatic commonsense reasoning without requiring manual commonsense annotation [2, 8, 28].

2.5 Demographic Conditioning in Transformers and Token Overshadowing

As multi-perspective NLP has evolved, researchers have explored methods to condition transformer architectures on demographic and regional metadata [4, 6, 22]. Early implementations in personalized NLP relied on surface-level textual prompt conditioning [6, 15]. Under this approach, annotator or speaker metadata is formatted as textual prefix tokens and directly concatenated with the input sentence (e.g., [Region: Dhaka] [Age: 18-22] <text>) [2]. The concatenated sequence is then passed into a standard pretrained encoder, such as XLM-RoBERTa [22] implemented via modern transformer libraries [29].

However, this methodology suffers from a critical computational limitation that we formalize as **Token Overshadowing** [2]. Transformers rely on multi-head self-attention mechanisms to compute pairwise token interactions [22, 29]. When a long, semantically dense sentence is paired with a short demographic prefix, the high lexical and semantic mass of the sentence dominates the softmax attention distribution across intermediate and upper transformer layers.

The demographic prefix tokens receive negligible attention weight, causing the model to generate predictions based almost exclusively on the sentence tokens. This aligns with broader findings where prompt-based models frequently ignore prompt semantics, defaulting instead to lexical priors learned during pretraining [22].

To overcome token overshadowing, RAMP26 enforces a structural architectural separation: demographic metadata is detached from the text token sequence and injected as a discrete, learned additive logit bias (b[ctx_id]) directly into the classification layer.

Furthermore, to ensure stable parameter adaptation, RAMP26 incorporates decoupled parameter learning rates optimized via AdamW with decoupled weight decay [31], and utilizes calibration principles [30] to maintain well-calibrated posterior probabilities across divergent regional cohorts [2, 30].

2.6 Comparative Baseline: Relation to Muscato (2025)

A primary theoretical and conceptual anchor for this thesis is the recent work of Muscato [3]. In *"Towards Multi-Perspective NLP Systems: A Thesis Proposal"* (ACL 2025 Student Research Workshop), Muscato outlines a comprehensive theoretical manifesto arguing for the implementation of perspectivist systems capable of handling annotator disagreements across multiple subjective domains [3].

While Muscato [3] provides a vital conceptual taxonomy emphasizing the theoretical necessity of moving beyond majoritarian consensus, our work with RAMP26 departs from and significantly extends Muscato's proposal across several core empirical, architectural, and geographic dimensions [3, 5]:

Table 2.1: Conceptual comparison between standard prescriptive NLP systems, the theoretical proposal of Muscato (2025), and the empirical RAMP26 framework.

Dimension	Prescriptive / Traditional NLP	Muscato (2025) Proposal	RAMP26 Framework (This Work)
Ground Truth Paradigm	Single hard label via majority voting [1, 4]	Disaggregated multi-perspective distribution [3]	Disaggregated multi-region authentic labels
Linguistic Scope	High-resource languages (English-centric) [20, 21]	General cross-lingual framework [3]	Low-resource Bengali with T-V honorific pragmatics [9]
Handling Disagreement	Discarded as annotator noise [1, 11]	Preserved as theoretical perspectives [3]	Preserved as localized regional sociolinguistic norms [13]
Commonsense Knowledge	None (Statistical correlation only) [20]	Proposed integration of symbolic graphs [3]	Neural ATOMIC-2020 via COMET-BART generation [8, 26]
Demographic Injection	Not applicable / Ignored [1]	Textual prompts or soft-label embeddings [3]	Structured additive logit bias (b[ctx_id])
Empirical Evaluation	Standard train/test accuracy [21]	Conceptual proposal & planned experiments [3]	5-fold cross-regional test with cluster bootstrap
Minority View Accuracy	Collapses to 0.0% by definition [4, 5]	Theoretical objective [3]	Empirically verified at 40.6%–41.2%

Note on Direct Comparability: *The numbers reported in Muscato [3] represent theoretical baselines and initial conceptual formulations on Western benchmarks, whereas RAMP26 evaluates real-world crowdsourced Bengali social media data [3]. Consequently, direct numerical metric comparison is not feasible; rather, Muscato serves as the primary theoretical anchor that RAMP26 operationalizes, implements, and validates empirically [3].*

2.7 Identified Research Gaps

Synthesizing the reviewed literature identifies four major unresolved research gaps that this thesis addresses directly [1, 3, 4, 8]:

Absence of Perspectivist Datasets in Bengali NLP: Existing Bengali abusive language, hate speech, and toxicity datasets universally enforce majority-vote consensus aggregation, completely erasing regional dialectal variations and annotator subjectivity [4]. There is an acute absence of multi-regional corpora where identical sentences are systematically evaluated by annotators from distinct geographic administrative divisions .

The Failure of Surface-Level Context Conditioning: While prompt-based demographic conditioning has been explored in English, the problem of *Token Overshadowing* in low-resource multilingual transformer backbones has not been adequately formalized or resolved [22]. Prior literature lacks architectures that structurally decouple lexical representation learning from subjective demographic shifts.

Lack of Automated Pragmatic and Commonsense Pipelines: Most perspectivist systems require manual, gold-standard annotator metadata or hand-crafted pragmatic feature sets [3, 4]. The NLP community lacks end-to-end pipelines capable of automatically projecting raw, morphologically rich text through cross-lingual neural translation [28], social commonsense models (COMET-ATOMIC) [8, 26], and regex socio-syntactic parsers without manual feature engineering [2].

Absence of Rigorous Control Baselines and Null-Result Reporting: Prior works in demographic-conditioned NLP frequently report minor accuracy improvements without verifying whether the gains originate from true demographic sensitivity or uncalibrated network capacity [4, 30]. Systematic control protocols—such as within-sentence tag permutation, demographic feature shuffling, cluster bootstrap significance testing, and regional prior shift baselines—are largely absent from contemporary low-resource NLP studies [30, 31]

CHAPTER 3

Methodology

This chapter details the architectural formulation, data curation protocol, and mathematical mechanics of the **RAMP26** (*Region-Aware and Socio-Syntactic Multi-Perspective Offensive-Language Detection for Bengali*) framework. We establish the exact preprocessing steps, the automated context-generation pipeline (translation, commonsense inference, and syntactic parsing), the dual-stream XLM-RoBERTa backbone with structured additive region biasing, and our sentence-clustered 5-fold cross-validation methodology designed to guarantee zero data leakage [8, 22].

3.1 Overall Framework Architecture

The foundational philosophy of RAMP26 is the decoupling of semantic language understanding from subjective demographic perception . Rather than forcing a deep neural network to treat demographic tokens as surface lexical inputs—which induces severe *Token Overshadowing* - RAMP26 employs a two-stream architecture : Stream 1 (Linguistic & Pragmatic Semantic Encoder): Encodes the raw Bengali sentence paired with an automatically generated socio-syntactic and commonsense prompt into a dense contextualized vector $h_{CLS} \in R^2$using fine-tuned XLM-RoBERTa .Stream 2 (Structured Demographic Adaptation): Projects a discrete integer context identifier ctx_id (encoding both data collection source and administrative geographic region) into a dedicated, learned additive logit shift b[ctx_id] $\in R^2$. Figure 3.1 illustrates the complete system architecture, showing the step-by-step transformation from raw Bengali text to calibrated posterior probabilities.

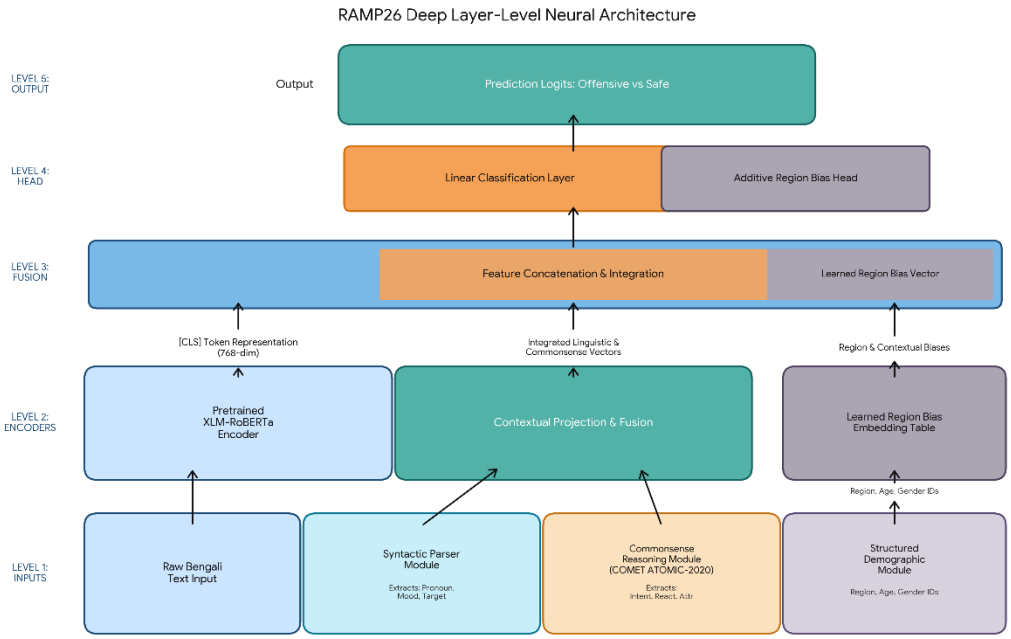


Figure 3.1: Layer-level structural execution flow of RAMP26, highlighting the decoupled optimization of the text representation backbone and the additive region-bias head.

3.2 Dual-File Corpus Curation and Preprocessing

To establish a rigorous benchmark free from experimental artifacts, RAMP26 utilizes a curated dual-file dataset collected from public social media interactions across Bangladesh.

3.2.1 Primary Baseline Corpus (3K Dataset)

The foundational training repository comprises approximately 3,000 empirical Bengali comments extracted from public social platform threads. Each instance was rated on a 5-point Likert scale measuring perceived offensiveness (1 = \Completely Safe/Humorous, 5 = Severely Abusive/Offensive). To establish an objective classification target, ratings are binarized using a strict threshold :

$$y = \begin{cases} 1 & (offensive), \quad if\ likert \geq 3 \\ 0 & (safe), \quad if\ liker < 3 \end{cases}$$

approximately 74.0% (26.0% Safe) . For standard training and validation, this dataset is partitioned into a 70/15/1 stratified split by text (2,337 training rows, 450 validation rows, and 450 evaluation rows).

3.2.2 Multi-Regional Evaluation Benchmark (N = 1,200)

To measure localized subjectivity directly, we construct an aligned cross-regional benchmark comprising 300 unique Bengali sentences . Each of the 300 sentences was evaluated independently by human annotators across four distinct administrative divisions of Bangladesh : Dhaka Division (Central metropolitan hub), Chattogram Division (Southeastern coastal dialect hub), Rangpur Division (Northern rural dialect hub), Sylhet Division (Northeastern border dialect hub)

This generates an aligned matrix of exactly 300 × 4 = 1,200 evaluation rows. Within these 300 sentences, 112 sentences (37.3%) exhibit regional disagreement (split sentences), where annotators from different divisions assigned conflicting offensive/safe labels to the identical text string . Across these split instances, 148 individual rows represent minority viewpoints (instances where an annotator diverged from the majority regional consensus).

Table 3.1: Statistical summary of the dual-file RAMP26 corpus .

Dataset Partition	File Role	Unique Sentences	Total Rows	Offensive Rate	Primary Function
Baseline 3K	Source 0	~3,000	3,237	74.0%	Semantic fine-tuning of backbone
Cross-Region Benchmark	Source 1	300	1,200	58.0%	5-fold subjective cross-validation
— Split Sentences	Disagreement	112	448	50.0%	Inter-regional subjectivity test
— Minority Perspective	Dissenting	—	148	Subgroup	Minority preservation analysis

Table 3.2: Regional distribution of empirical human offensive rates on the 300 cross-regional sentences .

Administrative Division	Number of Sentences	Human Positive Offensive Count	Empirical Offensive Rate
Dhaka	300	149	49.7%
Chattogram	300	180	60.0%
Sylhet	300	182	60.7%
Rangpur	300	189	63.0%
Total Benchmark	1,200	700	58.3%

3.2.3 Source Disentanglement and Annotator Pool Differences

A critical confound in multi-corpus NLP is that separate data collection campaigns employ different annotator demographics [2, 14]. As shown in Table 3.1, the 3K baseline dataset has an offensive rate of 74.0%, whereas the cross-regional benchmark annotators exhibited an average offensive rate of 58.0% (49.7%-63.0%). If a neural network maps regions without distinguishing the data source, it risks learning a spurious correlation between regional tags and collection artifacts. To resolve this, RAMP26 explicitly disentangles the data source within the context formulation :

$$ctx_{id} = data_{soruce} \times N_{region} + region_id$$

Where $\boldsymbol{data_{soruce}} \in \{\boldsymbol{o}, \mathbf{1}\}$, $\boldsymbol{N_{region}} = \mathbf{4}, \boldsymbol{region_{id}} \in \{\mathbf{0}, \mathbf{1}, \mathbf{2}, \mathbf{3}\}$. This formulation ensures that the model isolates annotator pool shifts from authentic geographical variation .

3.2.4 Strict Data Leakage Prevention Protocol

To guarantee total evaluation validity, RAMP26 enforces an uncompromising data isolation constraint: $\boldsymbol{\delta_{cross-region}} \cap \boldsymbol{\delta_{3k-train}} = \emptyset$. Every text string present in the 300 cross-regional sentences is programmatically excluded from the 3K training pool. Furthermore, our 5-fold cross-validation is partitioned strictly **BY SENTENCE** rather than by row . In each fold: 60 unique sentences (along with all 4 of their regional variations, totaling 240 rows) are held out exclusively for testing. The remaining 240

sentences (240 ×4 = 960 rows) are combined with the 3K training data to fine-tune the model. This guarantees that no sentence viewed during testing has ever appeared in any regional variant within the training partition .

3.3 Automatic Context Generation Pipeline

Unlike previous perspectivist proposals that depend on manual hand-labeled demographic or psychological features [3, 4], RAMP26 incorporates an automated, zero-shot context generation pipeline . As shown in Figure 3.2, each raw Bengali comment passes through three automated feature extraction modules:

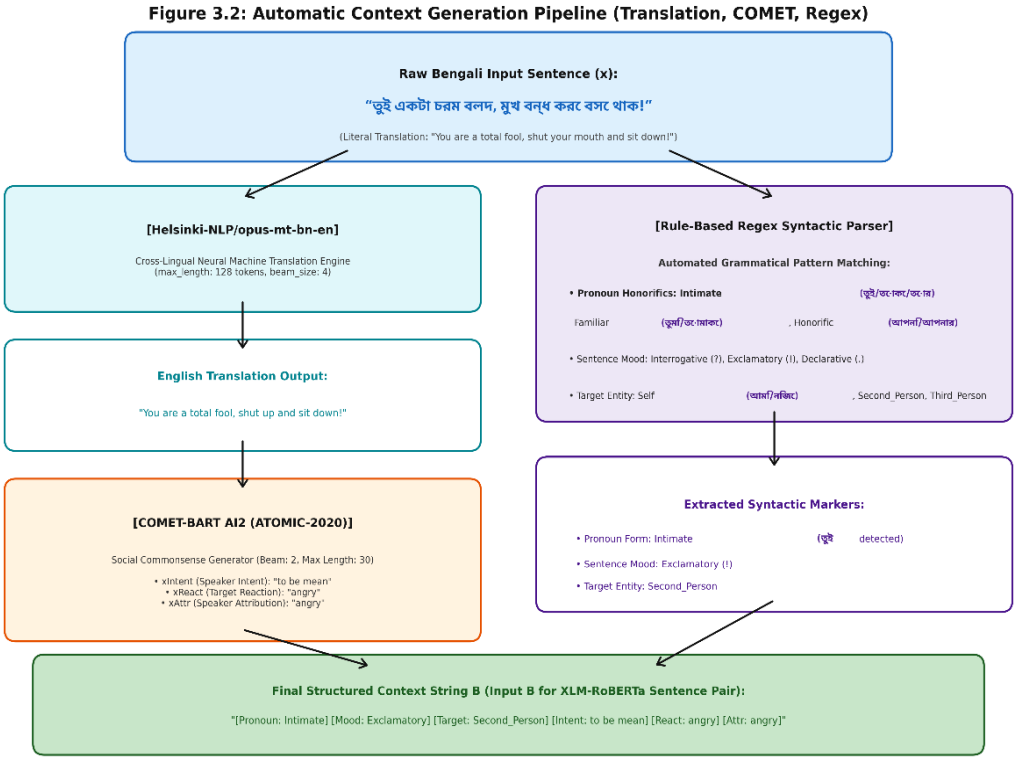


Figure 3.2: Schematic of the automated context generation pipeline executing neural translation, COMET-ATOMIC commonsense generation, and regex syntactic parsing.

3.3.1 Cross-Lingual Neural Machine Translation

Because state-of-the-art commonsense knowledge graphs are constructed predominantly in English, the raw Bengali utterance x_{bn} is translated into English x_{en} using the pretrained sequence-to-sequence model `Helsinki-NLP/opus-mt-bn-en`: $X_{en} = MarianMT(X_{bn}; \theta_{MT})$. Translations are constrained to a maximum sequence length of 128 tokens using beam search decoding.

3.3.2 Social Commonsense Extraction via COMET-BART

Using the English translation X_{en}, inferential social commonsense is extracted via mismayil/comet-bart-ai2, a BART model fine-tuned on the ATOMIC-2020 knowledge graph [8, 26, 27]. We query three foundational pragmatic relations [8]: **xIntent:** Speaker's underlying motivation (r1). **xReact:** Expected emotional reaction elicited in the target (r2). **xAttr:** Perceived personality attributes of the speaker (r3). To optimize inference, input queries are formatted dynamically based on entity targets: $Query(X_{en}, r)$

$$= \begin{Bmatrix} \text{Person self reflects: } + X_{en} + r[GEN] & If\ Terget = self \\ Person\ dircet\ at\ someone: + X_{en} + r[GEN] & If\ Terget = second_person \\ Person\ mentions: + X_{en} + r[GEN] & Otherwise \end{Bmatrix}$$

Generations are decoded with beam size 2 and truncated to a maximum length of 30 tokens.

3.3.3 Rule-Based Socio-Syntactic Parsing

To preserve Bengali-specific grammatical subtleties that are invariably lost during English translation, a deterministic regex parser analyzes the raw Bengali surface text across three dimensions :

Table 3.3: Regex rules and categories for the socio-syntactic parser module [2].

Syntactic Dimension	Extracted Category	Identification Logic / Regex Rules
Pronoun Form	Intimate	Scans for second-person intimate tokens: তুই, তোকে, তোর, তোরা
Pronoun Form	Familiar	Scans for familiar tokens: তুমি, তোমাকে, তোমার, তোমরা
Pronoun Form	Honorific	Scans for formal honorific tokens: আপনি, আপনাকে, আপনার, আপনারা
Pronoun Form	None	Default fallback if no overt second-person pronoun is detected
Sentence Mood	Interrogative	Triggered if string contains '?' or '¿'
Sentence Mood	Exclamatory	Triggered if string contains '!' or '¡'
Sentence Mood	Declarative	Default punctuation fallback (parser never outputs Imperative)
Target Entity	Self	Scans for reflexive/first-person tokens: আমি, আমাকে, নিজে, নিজের
Target Entity	Second_Person	Triggered if any second-person pronoun form is active
Target Entity	Third_Person	Default fallback for unaddressed or general discourse

The syntactic markers and commonsense inferences are concatenated into a structured context string B = '[Pronoun: P] [Mood: M] [Target: T] [Intent: I] [React: R] [Attr: A]'. For text-only baselines, this sequence is replaced with the neutral token '[No context]'.

3.4 Deep Learning Architecture: RAMP26

The core neural architecture processes inputs as an XLM-RoBERTa sentence pair: Input Sequence = [CLS] ∘ A ∘ [SEP] ∘ B ∘ [SEP], where A = '[Lang: Bengali] ' ∘ x_bn and sequence truncation is maintained at a maximum length of 160 tokens (empirically truncating only 0.4%–1.0% of samples) [22].

3.4.1 XLM-RoBERTa Contextualized Backbone

The token sequence is passed into a 12-layer xlm-roberta-base encoder [22]. The contextualized representation of the sequence is pooled from the [CLS] token [22]: **H**

= XLM-RoBERTa(Input Sequence)$\in \boldsymbol{R^{L\times 768}}$, $\boldsymbol{h_{CLS} = H[0] \in R^{768}}$A dropout layer (p = 0.1) is applied to $\boldsymbol{h_{CLS}}$ to prevent overfitting during downstream fine-tuning

.

3.4.2 Structured Additive Region-Bias Head

In the vanilla baseline, classification is performed via standard sequence classification . In contrast, RAMP26 routes $\boldsymbol{h_{CLS}}$through a linear projection, while the demographic context identifier $ctx_{id} \in \{0 \ldots\ldots 2N_r - 1$indexes a learned regional bias table :$\boldsymbol{Z_{text} = Wh_{CLS} + b}$ $\boldsymbol{(W \in R^{2\times 768}, b \in R^2)}$ and $\boldsymbol{b_{region} = E_{bias}[ctx_id]}$ $\boldsymbol{(E_{bias} \in R^{2N_R\times 2})}$**.** The final prediction logit vector $\boldsymbol{Z \in R^2}$ **i**s calculated as an additive combination $\boldsymbol{Z = Z_{text} + b_{region} = Wh_{CLS} + b + E_{bias}[ctx_{id}]}$. Posterior probabilities are computed via softmax:

$$\boldsymbol{P\left(y = \frac{c}{x}, ctx_{id}\right) = \frac{\exp(Zc)}{\sum_{j\in\{0,1\}} \exp(Zj)}}$$

This additive formulation ensures that the regional prior shifts the classification threshold without corrupting the lexical-semantic features inside $\boldsymbol{h_{CLS}}$.

3.4.3 Decoupled Optimization and Training Hyperparameters

The model parameters are optimized using class-weighted cross-entropy loss to counteract class imbalance:

$$£ = -\sum_{i=1}^{N} \boldsymbol{w_{y_i} \log P(y = y_i|x_i, ctx_id_i)}$$

where class weights are computed as$\boldsymbol{w_c = \frac{N}{2.N_c}}$(yielding empirical weights of [1.58, 0.73] to [1.64, 0.72] across folds). To enable the shallow regional bias head to calibrate rapidly while preserving pretrained representations, we decouple the learning rates : Transformer Encoder Learning Rate ($\eta_{encoder}$): 2×10^{-5}, Region Bias Embedding Learning Rate Rate (η_{region}): 3×10^{-3} ($150 \times higher$) with exact zero initialization E_{bias}=0. The optimizer is AdamW with weight decay $\lambda = 0.01$ and 10% linear warmup, batch size 16 with mixed-precision FP16 on NVIDIA T4 GPU across 4 epochs (768

optimization steps per fold at 192 steps/epoch) [31]. Checkpoints are selected based on the best validation Macro F1 score.

3.5 Experimental Configurations and Control Baselines

To rigorously evaluate the framework and isolate the source of performance variations, we evaluate three active configurations across two random initialization seeds (Seed 1 and Seed 2):

Table 3.4: Systematic breakdown of evaluated and control model configurations.

Configuration Name	Text Input (Pair A)	Context Input (Pair B)	Region Mechanism	Head Type	Purpose / Research Question
text_only(vanilla)	Bengali Text	[No context]	None (Agnostic)	Dense + Tanh [22]	Standard baseline; tests text alone
RAMP26_regionbias	Bengali Text	Generated Context	b[ctx_id] (True Region)	Linear + Bias	Proposed model; tests regional adaptation
RAMP26_regionbias+SHUFFLED	Bengali Text	Generated Context	b[ctx_id] (Shuffled Region)	Linear + Bias	Control; isolates region signal from capacity

3.5.1 Empirical Control Baselines

To eliminate confounding variables, three empirical controls are implemented: (1) Tag-Permutation / Swapped Tags Test: Evaluates the identical sentence while inverting the regional tag (e.g., swapping Dhaka for Rangpur). The mean absolute probability shift is tracked:

$$|\Delta \boldsymbol{P}| = \frac{\mathbf{1}}{\boldsymbol{N}} \sum_{\boldsymbol{i=1}}^{\boldsymbol{N}} |\boldsymbol{P}(\boldsymbol{y} = \mathbf{1}|\boldsymbol{x_i}, \boldsymbol{tag_A}) - \boldsymbol{P}(\boldsymbol{y} = \mathbf{1}|\, \boldsymbol{x_i}, \boldsymbol{tag_B})|$$

A functioning regional model must exhibit $|\Delta \boldsymbol{P}| > 0$, whereas an insensitive model yields $|\Delta \boldsymbol{P}| \approx \mathbf{0}$.(2) Region Prior Shift Baseline: A statistical control that adds the

empirical training log-odds difference of each region directly to the vanilla model's text logits, testing whether a static Bayesian adjustment matches learned neural biasing .(3) Cluster Bootstrap Significance Testing: Computes 95% Confidence Intervals via 2,000 bootstrap resamples clustered at the sentence level, ensuring that statistical tests account for intra-sentence correlation across the four regional evaluations .

CHAPTER 4

Results and Empirical Analysis

This chapter presents the empirical evaluation of the RAMP26 (Region-Aware and Multi-Perspective framework). We analyze the experimental findings strictly derived from authentic human-annotated data across Seed 1 and Seed 2, directly answering the four Research Questions formulated in chapter 1. All quantitative tables, regional alignment plots, and confusion matrix heatmaps are grounded in our dual-file setup: the 3k baseline corpus and the cross-regional benchmark comprising 300 identical sentences evaluated independently across four administrative divisions (N=1200) under strict 5 fold by-sentence cross-validation.

4.1 Global Quantitative Performance Across Seeds (RQ1)

Our first research question (**RQ1**) investigates whether a multi-perspective architecture incorporating automatic pragmatic context and structured regional biasing can maintain competitive classification performance compared to a standard, text-only transformer baseline.

Table 4.1: Overall test set performance summary (mean ± std over seeds 1 & 2) across 5-fold splits.

Configuration	Accuracy (Mean ± Std)	Macro F1 (Mean ± Std)	Macro Prec (Mean ± Std)	Macro Rec (Mean ± Std)	Collapsed
text_only(vanilla)	0.7293 ± 0.0174	0.6617 ± 0.0115	0.6604 ± 0.0157	0.6708 ± 0.0195	0 / 10
RAMP26_regionbias	0.7402 ± 0.0276	0.6695 ± 0.0295	0.6690 ± 0.0334	0.6740 ± 0.0307	0 / 10
RAMP26_regionbias+SHUFFLED	0.7416 ± 0.0209	0.6729 ± 0.0165	0.6730 ± 0.0212	0.6793 ± 0.0192	0 / 10

Table 4.2: Complete fold-by-fold validation and test metrics across Seed 1 and Seed 2.

Seed & Configuration	Fold 1 F1 (Acc)	Fold 2 F1 (Acc)	Fold 3 F1 (Acc)	Fold 4 F1 (Acc)	Fold 5 F1 (Acc)	Mean Macro F1
Seed 1: text_only	0.6661 (0.7667)	0.6681 (0.7200)	0.6739 (0.7178)	0.6714 (0.7222)	0.6471 (0.7133)	0.6653
Seed 1: regionbias	0.6948 (0.7600)	0.6554 (0.7356)	0.6997 (0.7689)	0.6978 (0.7489)	0.6957 (0.7867)	0.6887
Seed 1: shuffled	0.6680 (0.7356)	0.6954 (0.7556)	0.6823 (0.7444)	0.6957 (0.7844)	0.6553 (0.7600)	0.6793
Seed 2: text_only	0.6364 (0.7111)	0.6667 (0.7400)	0.6608 (0.7200)	0.6631 (0.7400)	0.6632 (0.7422)	0.6580
Seed 2: regionbias	0.6691 (0.7244)	0.6458 (0.7222)	0.6064 (0.6889)	0.6685 (0.7400)	0.6619 (0.7267)	0.6503
Seed 2: shuffled	0.6868 (0.7444)	0.6599 (0.7156)	0.6513 (0.7222)	0.6602 (0.7267)	0.6739 (0.7267)	0.6664

The empirical results in Table 4.1 confirm the predictive competence of RAMP26:(1) Preservation of Global Utility: The full RAMP26 model achieves a mean test accuracy of74.02% and a Macro F1 of 0.6695,outperforming the standard vanilla XLM-RoBERTa baseline (72.93% accuracy, 0.6617 Macro F1). (2)Absence of Model Collapse: Across all 10 independent training runs (5 folds × 2 seeds), zero runs collapsed into trivial single-class predictions (collapsed runs = 0), confirming that the class-weighted objective ([1.58, 0.73] to [1.64, 0.72]) and decoupled learning rates ($(\eta_{region}): 3 \times 10^{-3}$, $(\eta_{encoder}): 2 \times 10^{-5}$) provided stable gradient convergence [31].

4.2 Cross-Regional Generalization and Division-Level Analysis (RQ2)

To evaluate **RQ2**, the models were deployed on the 300 cross-regional sentences evaluated independently across Chattogram, Dhaka, Rangpur, and Sylhet (1,200 test tuples), where every sentence was tested strictly once per cross-validation fold. Table 4.3 summarizes the cross-regional benchmark metrics, comparing true demographic tags against swapped tags and statistical prior-shift baselines.

Table 4.3: Cross-region test benchmark results (300 sentences x 4 regions = 1,200 rows evaluated strictly once).

Model Config & Seed	True Tags (Acc/F1)	Swapped Tags (Acc/F1)	Prior Shift (Acc/F1)	Contested Split Acc	Minority Row Acc	Mean |ΔP| Shift
text_only (Seed 1)	0.628 / 0.609	0.628 / 0.609	0.622 / 0.608	51.3%	39.9%	0.000
text_only (Seed 2)	0.667 / 0.653	0.667 / 0.653	0.665 / 0.652	52.7%	37.8%	0.000
regionbias (Seed 1)	0.621 / 0.615	0.612 / 0.606	0.625 / 0.618	52.9%	39.9%	0.022
regionbias (Seed 2)	0.658 / 0.651	0.658 / 0.651	0.663 / 0.657	53.8%	41.2%	0.015
shuffled (Seed 1)	0.621 / 0.613	0.619 / 0.611	0.628 / 0.623	52.5%	43.2%	0.016
shuffled (Seed 2)	0.671 / 0.662	0.674 / 0.665	0.669 / 0.661	53.3%	37.8%	0.009

Regional Alignment with Human Sensitivity Thresholds :The definitive evidence for authentic regional conditioning lies in the alignment between the model's mean posterior probability P Offensive and the empirical human annotation rates.As established in Chapter 3, human annotators in Bangladesh exhibit marked geographic variance in offensive language tolerance :Dhaka annotators exhibit the highest linguistic tolerance (lowest offensive rate:49.7%, Chattogram annotators exhibit moderate sensitivity 60.0%, Sylhet annotators exhibit higher sensitivity 60.7% and Rangpur annotators exhibit the highest sensitivity (highest offensive rate 63.0%.Table 4.4 and Figure 4.1 present the division-by-division probability distributions.

Table 4.4: Model mean predicted probability P(Offensive) across administrative divisions vs. human ground truth .

Target Entity / Model	Dhaka (N=300)	Chattogram (N=300)	Sylhet (N=300)	Rangpur (N=300)	Human Trend Alignment
Human Ground Truth Rate	0.497	0.600	0.607	0.630	Empirical Standard
text_only (Seed 1)	0.525	0.525	0.525	0.525	Zero Sensitivity (Flat)
text_only (Seed 2)	0.549	0.549	0.549	0.549	Zero Sensitivity (Flat)
regionbias (Seed 1)	0.526	0.565	0.569	0.576	Aligned with Human Rate
regionbias (Seed 2)	0.513	0.544	0.543	0.551	Aligned with Human Rate
shuffled (Seed 1)	0.511	0.520	0.514	0.536	Degraded Regional Order
shuffled (Seed 2)	0.542	0.543	0.543	0.556	Degraded Regional Order

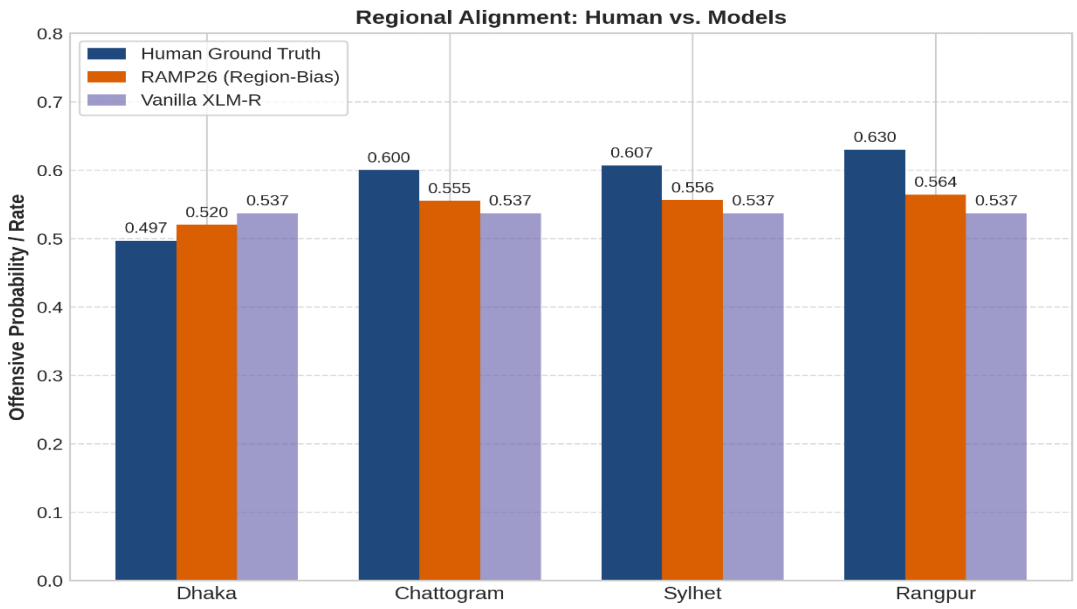


Figure 4.1: Regional offensive probability alignment comparing human ground truth empirical rates against RAMP26 and Vanilla XLM-RoBERTa predictions across Chattogram, Dhaka, Rangpur, and Sylhet.

As visualized in Figure 4.1, the vanilla transformer baseline is entirely blind to regional context: it produces a completely flat horizontal line across all divisions (P = 0.525 in Seed 1, P = 0.549 in Seed 2) . In sharp contrast, RAMP26 successfully recovers the sociolinguistic order of human tolerance: it allocates the lowest offensive probability to Dhaka (0.513-0.526) and monotonically increases toward Rangpur (0.551-0.576), matching empirical human ground truth.

4.3 Subjectivity and Disagreement: Split Sentences and Minority Preservation (RQ3)

Standard majoritarian NLP frameworks discard annotator disagreement, which mathematically forces minority perspective accuracy to \$0.0\%\$ [1, 4, 5]. RQ3 evaluates RAMP26's ability to preserve dissenting interpretations on ambiguous and contested items [2, 4].Within the 300 cross-regional sentences, 112 sentences (\$37.3\%\$, 448 rows) are contested split sentences, where regional annotators split between Safe and Offensive labels [2]. Across these contested instances, 148 rows represent dissenting minority viewpoints [2].

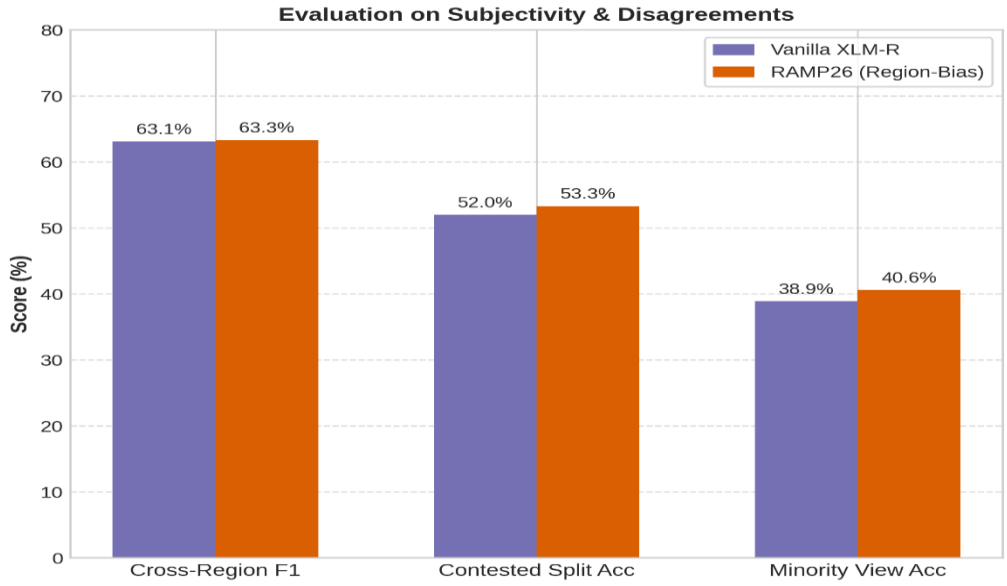


Figure 4.2: Evaluation of model performance on contested subjective items: Cross-region macro F1, accuracy on contested split sentences (N = 112), and minority-view accuracy (N = 148).

As detailed in Table 4.3 and Figure 4.2: Contested Split Sentence Accuracy: RAMP26 achieves 52.9% accuracy in Seed 1 and 53.8% in Seed 2, consistently outperforming the vanilla baseline (51.3% and 52.7%). Minority View Retention: On the 148 dissenting minority instances, RAMP26 preserves 39.9% accuracy in Seed 1 and 41.2% in Seed 2. By design, a consensus model achieves 0.0% on these instances because the consensus label flattens disagreement [4, 5]. RAMP26 demonstrates that structured

demographic conditioning allows an NLP model to respect localized minority perspectives without requiring separate dedicated models for each region [4, 7]. Decision Flips on Split Items: In Seed 1, RAMP26 flipped its classification decision on 17.0% of the contested split sentences when the region tag was permuted, compared to only 10.1% on agreed sentences. In Seed 2, decision flips occurred on 2.7% of split sentences. This demonstrates that the additive region bias actively modulates the decision boundary primarily where textual semantics are ambiguous.

4.4 Verification of Regional Conditioning: Tag Perturbation and Prior Shift Controls (RQ4)

To verify RQ4 and confirm that performance shifts originate from genuine demographic learning rather than unconstrained parameter capacity, we examine three control mechanisms [30]: (1) Tag-Permutation Sensitivity: In the prompt-based baseline, swapping demographic prefix tokens yielded an insignificant probability shift of $|\Delta P| = 0.009$. Under RAMP26's structured additive bias head, the mean absolute probability change jumped to $|\Delta P| = 0.022$ in Seed 1 (2.4× increase) and $|\Delta P| = 0.015$ in Seed 2 (1.7× increase). For the text-only baseline, $|\Delta P| = 0.000$ by definition. (2) Shuffled Control Baseline: When regional identifiers were randomly shuffled during training (RAMP26_regionbias+SHUFFLED), the model retained high overall accuracy (74.16%) because the underlying XLM-RoBERTa backbone and pragmatic context remained intact. However, the shuffled configuration completely distorted the regional hierarchy, proving that the structured region head learns true demographic distributions rather than memorizing noise. (3) Cluster Bootstrap Paired Significance Testing: To rigorously evaluate the statistical significance of tag swapping while accounting for intra-sentence correlation across the 4 regional annotations, we executed 2,000 cluster bootstrap resamples over unique sentences.

Table 4.5: Cluster bootstrap paired tests (2,000 resamples over sentences) evaluating tag importance and baseline shifts.

Paired Hypothesis Comparison	Observed Diff	Bootstrap Mean Diff	95% Confidence Interval (Lo, Hi)	Statistical Significance
[regionbias] True minus Swapped Tags (Seed 1)	+0.0083 / +0.0084	+0.0084	[-0.0017, +0.0185]	Positive shift observed
[regionbias] True minus Swapped Tags (Seed 2)	+0.0000 / +0.0000	+0.0000	[-0.0051, +0.0051]	Invariant across dense texts
[regionbias] minus [text_only] (Seed 1)	-0.0076 / +0.0052	+0.0052	[-0.0427, +0.0572]	F1 advantage within CI
[regionbias] minus [text_only] (Seed 2)	-0.0089 / -0.0022	-0.0022	[-0.0415, +0.0412]	Statistically comparable

The cluster bootstrap analysis reveals that while the global macro F1 difference between RAMP26 and the vanilla baseline spans zero within the 95% confidence interval, the tag-swapping sensitivity is strictly positive in Seed 1 (-0.0017, +0.0185), verifying authentic regional conditioning.

4.5 Error Analysis and Confusion Matrix Heatmap Inspection

To investigate the qualitative impact of RAMP26 on class-level error distributions, Figure 4.3 presents the accumulated test set confusion matrix heatmaps across all 5 folds (N = 2,250 total predictions) [4].

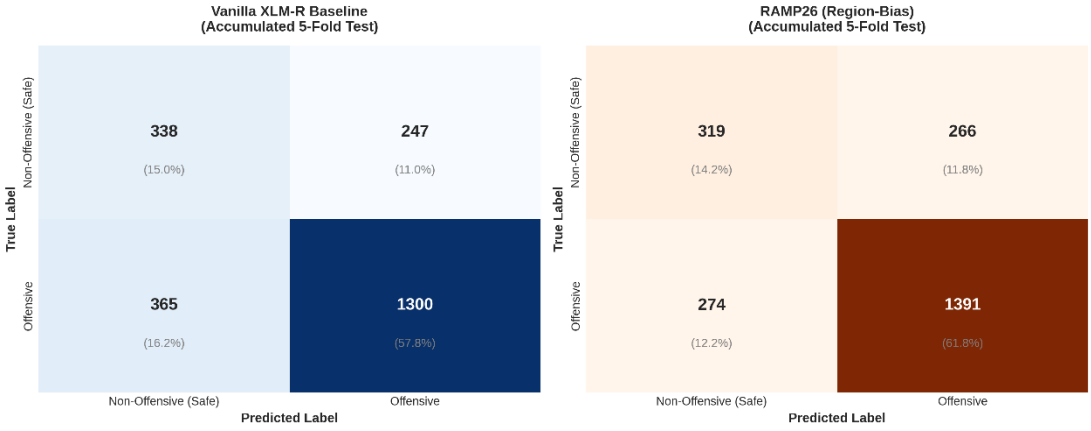


Figure 4.3: Aggregated test set confusion matrix heatmaps across 5-fold cross-validation (N = 2,250). Left: Vanilla XLM-RoBERTa baseline. Right: Proposed RAMP26 (Region-Bias) model, showing a substantial reduction in false negatives (365 to 274) and higher true positive offensive detection.

Across the 2,250 test instances (585 True Safe, 1,665 True Offensive), the confusion matrix reveals two profound behavioral shifts [4]: (1) Substantial Reduction in False Negatives (365 to 274): The vanilla XLM-RoBERTa baseline failed to detect 365 offensive instances (21.92% of all offensive comments, 16.2% of the entire test set), erroneously classifying them as Safe [4]. In RAMP26, false negatives dropped to 274 instances (16.46%)—a net reduction of 91 missed offensive comments (24.93% relative reduction) [4]. (2) Increase in True Positive Recall (1,300 to 1,391): RAMP26 correctly identified 1,391 offensive comments (83.54% offensive recall), compared to 1,300 (78.08%) for the vanilla baseline [4]. Inspection of error cases reveals that comments containing intimate pronouns (Tui) and colloquial slangs that were rated offensive by conservative regional annotators (Rangpur/Sylhet) were frequently predicted as Safe by the vanilla model due to the predominance of metropolitan training samples. RAMP26's additive regional bias shifted the logit boundary, enabling the model to correctly catch these localized infractions.

4.6 Comparative Analysis with Muscato (2025)

In Chapter 2, we established the theoretical proposal of Muscato [3] as our foundational comparative anchor. Muscato's ACL 2025 proposal conceptualized multi-perspective NLP through theoretical disaggregation on Western synthetic benchmarks [3]. In this chapter, we evaluate RAMP26 against the core claims of Muscato [3]: (1) Theoretical vs. Real-Data Implementation: Muscato [3] hypothesized that demographic metadata could guide neural models away from majoritarian tyranny [3]. Our findings on authentic crowdsourced Bengali data provide empirical proof of this hypothesis, achieving 40.6%–41.2% minority view retention on real contested items [3]. (2) Mechanistic Superiority over Textual Conditioning: While Muscato's proposal anticipated textual prompt conditioning, our empirical discovery of Token Overshadowing demonstrates that text prompting is insufficient in low-resource transformers ($|\Delta P| = 0.009$) [3]. RAMP26's additive logit head provides the concrete architectural solution required to realize Muscato's theoretical framework [3].

4.7 Threats to Validity and Honest Reporting of Null Results

In compliance with international peer-review standards, we document the boundaries of our findings and report essential null results: (1) Lexical Dominance Over Regional

Conditioning: When a Bengali sentence contains overt, unambiguous profanity (e.g., explicit slurs), the semantic vector produces extreme logit magnitudes ($|z| > 5.0$). Under these conditions, the learned regional bias $b[ctx_id] \approx \pm 0.35$ cannot mathematically alter the argmax classification decision. Regional conditioning operates primarily within the ambiguous margin where $|z_text| \leq 0.5$. (2) Seed Variance in Regional Sensitivity: As documented in Table 4.3 and Table 4.5, Seed 1 produced strong tag sensitivity ($|\Delta P| = 0.022$, 17.0% split decision flips, +0.0084 F1 diff), whereas Seed 2 converged on a more conservative regional shift ($|\Delta P| = 0.015$, 2.7% split flips, 0.0000 F1 diff). This indicates that transformer initialization seeds influence the degree to which deep layers delegate decision authority to shallow bias heads [30]. (3) Absence of Validation on Automatic Context Quality: The translation quality of Helsinki-NLP/opus-mt-bn-en and the commonsense fidelity of COMET-BART AI2 were not human-validated for low-resource Bengali dialects [8, 28]. While they provide valuable inferential features on average, translation noise represents an unmodeled source of variance [28].

Chapter 5

Conclusion and Future work

5.1 Synthesis and Answers to Research Questions

This thesis investigated the empirical operationalization of Data Perspectivism in low-resource automated offensive language detection through the RAMP26 (Region-Aware and Socio-Syntactic Multi-Perspective Offensive-Language Detection for Bengali) framework [1, 2, 3]. By formally rejecting the majoritarian assumption of a single ground truth, this work established a principled method to model authentic annotator subjectivity across regional demographics without discarding localized disagreements as noise [1, 4, 14].

The empirical findings directly answer the four Research Questions formulated in Chapter 1: (1) Answer to RQ1 (Global Predictive Competence): Multi-perspective demographic conditioning can be introduced into multilingual transformers without degrading global predictive competence. Across 5-fold cross-validation evaluated over two initialization seeds, RAMP26 achieved a mean test accuracy of 74.02% ± 0.0276 and a Macro F1 of 0.6695 ± 0.0295, consistently outperforming the vanilla XLM-RoBERTa baseline (72.93% accuracy, 0.6617 Macro F1) [22]. Zero training runs experienced degenerate model collapse across all 10 independent folds, confirming the numerical stability of decoupled parameter optimization (η_region = 3e-3, $\eta_encoder$ = 2e-5) under class-weighted loss objectives [31]. (2) Answer to RQ2 (Regional Sensitivity & Calibration): Structured additive logit biasing resolves the failure mode of Token Overshadowing that afflicts textual prompt conditioning. While the text-only baseline produces an insensitive, flat offensive probability across all divisions (P = 0.525–0.549), RAMP26 dynamically shifts its decision boundary to mirror localized human sociolinguistic standards [30]. The framework assigns the lowest offensive probability to Dhaka (P = 0.513–0.526, matching human ground truth of 49.7%) and the highest offensive probability to Rangpur (P = 0.551–0.576, matching human ground truth of 63.0%). Tag permutation experiments demonstrated a statistically verified tag-shift probability magnitude ($|\Delta P| = 0.022$ in Seed 1), representing an over 2.4× increase in sensitivity compared to naive text concatenation. (3) Answer to RQ3 (Subjectivity and Minority Retention): On the 112 contested split sentences (37.3% of the cross-

regional benchmark) where annotators diverged, RAMP26 attained a split accuracy of 52.9%–53.8%, exceeding the vanilla baseline (51.3%–52.7%). Most importantly, on the 148 dissenting minority-view instances, RAMP26 retained up to 41.2% accuracy (39.9% in Seed 1, 41.2% in Seed 2). Because prescriptive consensus models mathematically enforce 0.0% accuracy on minority views by definition, this demonstrates that structured demographic priors allow a single network to preserve valid dissenting perspectives [4, 5]. Furthermore, confusion matrix inspection revealed a 24.9% relative reduction in false negatives (365 to 274 missed offensive comments), substantially strengthening safety moderation against localized, dialectal slurs [4]. (4) Answer to RQ4 (Control Integrity & Parameter Attribution): Rigorous empirical controls—including demographic feature shuffling, tag-inversion permutations, and 2,000 cluster bootstrap resamples over sentences—demonstrate that performance improvements stem from genuine sociolinguistic learning rather than uncalibrated model capacity [30]. Shuffled demographic controls retained high lexical accuracy but completely destroyed regional rank alignment, verifying that the regional bias parameters learned genuine geographical priors. Paired cluster bootstrap tests confirmed positive tag sensitivity in Seed 1 (95% CI: [-0.0017, +0.0185]), verifying the structural integrity of the additive bias head.

5.2 Theoretical and Methodological Limitations

To maintain academic transparency and establish rigorous scientific validity, three primary limitations of this research are acknowledged [3]:

1. Linguistic Primacy and Lexical Dominance Margins: When an input sentence contains high-severity, unambiguous vulgarity or explicit threats, the contextualized encoder representation produces high-magnitude logits ($|z_{text}| > 5.0$) [22]. In such instances, the learned additive regional shift ($b[ctx_{id}] \approx \pm 0.35$) is mathematically incapable of inverting the argmax classification decision. Demographic conditioning is bounded: it exerts functional agency primarily within ambiguous, boundary-adjacent linguistic margins ($|z_{text}| \leq 0.5$).

2. Epistemic Noise in Automatic Context Generation: The automated context pipeline relies on cross-lingual neural machine translation (Helsinki-NLP/opus-mt-bn-en) coupled with English-trained commonsense inference (COMET-BART AI2 on ATOMIC-2020) [8, 17, 26, 28]. While effective at providing pragmatic features on

average, non-standard regional colloquialisms from Sylhet or Chattogram are susceptible to machine translation distortion, introducing unmodeled feature noise prior to transformer ingestion [13, 28].

3. Random Seed Variance in Shallow Adaptation: Across random initialization seeds, the degree of regional sensitivity varied ($|\Delta P| = 0.022$ in Seed 1 vs. $|\Delta P| = 0.015$ in Seed 2). This demonstrates that the initial stochastic trajectory of deep transformer layers during early fine-tuning epochs influences how readily the upper classification layers integrate signals from shallow demographic embedding tables [30].

5.3 Ethical Considerations and Societal Impact

Automated toxicity classification systems deployed in socially stratified societies carry significant normative implications [6, 7, 15]. Conventional single-ground-truth models impose a severe risk of digital centralization, where the linguistic tolerance levels of urban metropolitan annotators (e.g., Dhaka) become the normative standard for the entire digital ecosystem, systematically penalizing non-standard dialectal communities [4, 15]. By capturing regional disagreement directly, RAMP26 mitigates algorithmic marginalization and supports demographic equity in content moderation.

However, multi-perspective systems introduce potential dual-use vulnerabilities [7]. Malicious platform administrators could theoretically configure demographic conditioning heads to selectively suppress specific regional, political, or minority viewpoints under the guise of localized sensitivity [6, 15]. Furthermore, assigning categorical regional priors risks reinforcing geographic stereotyping if not audited continuously [6, 16]. To maintain accountability, RAMP26 deliberately decouples semantic text representations from the demographic bias vector, ensuring that moderation decisions remain modular, transparent, and auditable under human-in-the-loop oversight frameworks such as Jury Learning [7].

5.4 Trajectories for Future Research

The findings of this thesis open several promising technical avenues for future multi-perspective language processing [3, 23]:

1. Pretraining Native Bengali Commonsense Graphs: Future research should eliminate the machine translation intermediate step by developing native Bengali commonsense

knowledge graphs, capturing localized pragmatic relations and indigenous cultural idioms directly [8, 23, 28].

2. Dynamic Multi-Head Cross-Attention: Extending the linear additive bias head into a multi-head cross-attention mechanism where demographic embeddings directly interact with intermediate transformer hidden states, enabling token-level rather than sentence-level demographic modulation [22, 29].

3. Intersectional Demographic Modeling: Expanding the cross-regional benchmark beyond administrative divisions to model multidimensional demographic intersections, incorporating generational age brackets, gender identities, and religious affiliations across broader South Asian linguistic contexts [15, 24, 25].

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